

Worksheet 3 — Why One Variable at a Time Is Not Enough

STAT 481 — Multivariate Analysis • Lecture 3 • Sun, Aug 30, 2026

Take-home. Not collected, not graded — we work through it together at the start of Lecture 4.

1. Objective of this lecture

By the end of this sheet you should be able to:

1. standardise every column of a table at once, and say why standardising a *row* is the same arithmetic on the other margin;
2. run a one-variable-at-a-time screen at a stated threshold, report exactly which countries it flags, and say why no threshold rescues it;
3. build one number out of two standardised columns, *predict its standard deviation from r before computing it*, and use it to find what the screen missed.

2. The data, and how to get it in R

Data set: `worldhealth2022.csv`, in the course project — no package needed, `read.csv()` is base R.
Size: 176 countries $\times$ 8 indicators (World Bank, 2022). Open `STAT481.Rproj` first, so the relative path resolves. Nothing on this sheet needs a package you do not already have.

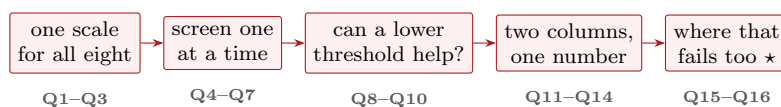
```
wh <- read.csv("data/worldhealth2022.csv", row.names = 1)
W  <- wh[, 4:11]           # the 8 indicators; cols 1-3 are labels
n  <- nrow(W); p <- ncol(W)
c(n = n, p = p)
```

<code>life_exp</code>	life expectancy at birth, years	<code>urban_pct</code>	urban population, %
<code>infant_mort</code>	infant mortality, per 1000 births	<code>gdp_pc</code>	GDP per capita, US\$
<code>fertility</code>	births per woman	<code>health_exp_pct</code>	health spending, % of GDP
<code>pop65</code>	population aged 65+, %	<code>internet_pct</code>	Internet users, %

3. What we want to know today

The question: Step 7 of today's lab asserts that Qatar is *unusual in no single variable, and unusual in the pair* — and never makes you check it. *Is that true, and if it is, what does it cost the standard way of hunting for unusual observations?*

Today: put all eight indicators on one scale, screen them one at a time and see exactly who that catches, find out whether a lower threshold rescues the method, and then build *one* number out of *two* columns and see what it finds.



4. The questions

Stage 1 — put all eight on one scale

`scale(W)` centres each column on its own mean and divides by its own SD (Worksheet 1 Q5; Lecture 2 Home Ex. 2b), so entry (i, j) is the **standard score** $z_{ij} = (x_{ij} - \bar{x}_j) / \sqrt{s_{jj}}$ — “how many of *this column's* SDs country i sits from *this column's* mean”. It is unitless, which is the point: years, dollars and percentages become comparable. It returns a *matrix*, so address it with `Z["QAT", ]`, never with `$`, and it works down **columns**.

Q1. Before running anything: what are the mean and the standard deviation of each of the eight columns of `Z <- scale(W)`? guess

Q2. Build `Z` and confirm your prediction on both statistics at once. R code

Q3. Read Qatar's row out of `Z` and check it against the eight numbers Step 7 printed. R code

Stage 2 — screen the variables one at a time

A one-variable-at-a-time screen at threshold k flags country i if *any* of its eight standard scores is far from zero: $\max_j |z_{ij}| > k$. It is the standard first pass on a new table — what `describe()`'s `min` and `max` invite you to do — and it treats the eight indicators as eight separate problems. Start at $k = 3$; `mx <- apply(abs(Z), 1, max)` holds each country's largest $|z|$, one number per country, so `MARGIN = 1` is right here for the first time today.

Q4. Before running it: at $k = 3$, how many of the 176 countries do you expect the screen to flag? guess

Q5. Compute each country's largest $|z|$, flag those above 3, and report how many and which. R code

Q6. Exactly one of those eight is on the list for a reason that has nothing to do with being an unusual country — find it, and name the variable. R code

Q7. Is Qatar flagged? Report its largest $|z|$ and its rank among the 176 on that measure. R code

Stage 3 — can a lower threshold rescue the screen?

Q8. To reach Qatar the threshold must fall below the value you found in Q7. Before running it: roughly how many countries will that flag? guess

Q9. Count the flags at $k = 3, 2.5, 2.4$ and 2 . R code

Q10. In one sentence: why does no choice of threshold rescue a one-variable-at-a-time screen here? guess

Stage 4 — two columns, one number

The contrast g . Today's Step 7 put life expectancy against the share aged 65+, because a country that lives a long time usually accumulates old people: those two columns correlate $r = 0.6596$. Define, for each country,

$$g_i = z_{i,\text{life_exp}} - z_{i,\text{pop65}},$$

one number per country, unitless, measuring **how much longer a country lives than its elderly share would lead you to expect**. Both columns are standardised, so g is a difference of two comparable things — which is what Stage 1 was for.

Q11. Both columns of Z have variance 1 and correlate $r = 0.6596$. Predict $\text{sd}(g)$ *exactly*, before computing it. guess

Q12. Build g , compute its standard deviation, and check it against your prediction. R code

Q13. Sort the countries on g and report the top six. R code

Q14. Look at the six countries you just listed — what does the *composition* of that list say about whether Qatar is an “outlier”? guess

Stage 5 — ★ where this fails too (run if you have time; this is the one to skip)

Q15. Where does the broken row of Q6 land on g ? Report the bottom three. R code

Q16. g used one pair because Step 7 pointed at it. How many pairs are there among the eight indicators — and what does that say about what we still need? guess

5. Conclusion — fill in the blanks

- Screening the eight indicators one at a time at $|z| > 3$ flags _____ countries. Qatar's largest standard score is _____, which ranks it _____ of 176 — so the screen _____ flag it.
- The first round threshold that reaches Qatar is $k =$ _____, and it flags _____ countries. Lowering the threshold buys more _____, never a better _____.
- With both columns standardised and $r = 0.6596$, $\text{sd}(g) = \sqrt{2(1-r)} =$ _____. On g , Qatar ranks _____ of 176, and the next five countries are _____.
- So a country can be unusual in _____ variable and still be the most unusual in the world in a _____ of them — and the only reason we found the right pair today is that _____.

6. What this opens

Every question here compared a country with the *average* country, and Stage 4 showed that doing so honestly means accounting for how two columns move together. Thursday we do it for all p columns at once, with one quantity that respects every correlation in **R** — so that “unusual” becomes something a computer can rank without being told which pair to look at. *Bring this sheet.*